Stereoisomerism

Cis–trans, E/Z and optical isomers

Name _____ Class _____ Date _____

KEY IDEAS

Cis–trans: C=C with two different groups on each carbon · E/Z: rank with CIP rules (Z = together, E = opposite)

CIP: higher atomic number first · tie: move outwards, first difference · C=O counts as (O, O) · R/S: point 4 away; 1 → 2 → 3 clockwise R, anticlockwise S

Chiral centre: carbon with four different groups · enantiomers: non-superimposable mirror images

Atomic numbers: H 1 · C 6 · O 8 · Cl 17 · Br 35

Draw the molecules when it helps. Mark chiral centres with an asterisk.

A. Types of isomer

1. For each pair, say whether they are structural isomers, cis–trans isomers or the same compound: (a) butane and methylpropane (b) cis- and trans-but-2-ene (c) propan-1-ol and propan-2-ol (d) two drawings of CH_3CH_2Cl rotated about the C–C bond

B. Cis–trans and E/Z

2. Which of these show cis–trans isomerism? Explain each. (a) pent-2-ene, $CH_3CH{=}CHCH_2CH_3$ (b) 2-methylpropene, $(CH_3)_2C{=}CH_2$ (c) 1,2-dibromoethene, $BrCH{=}CHBr$

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3. Draw the cis and trans isomers of 1,2-dichloroethene. Which has the higher boiling point (60 °C or 48 °C), and why?

4. Assign E or Z: (a) CHCl=CHBr with Cl and Br on the same side (b) $CH_3CH{=}CHCl$ with CH_3 and Cl on opposite sides

C. Optical isomerism

5. Identify any chiral centres in: (a) 2-chlorobutane, $CH_3CHClCH_2CH_3$ (b) propan-2-ol, $(CH_3)_2CHOH$ (c) lactic acid, $CH_3CH(OH)COOH$ (d) 1-chlorobutane

6. Describe how the two enantiomers of lactic acid are (a) the same (b) different. What is a racemic mixture?

D. Applications

7. Explain why a drug made as a single enantiomer can work better than the racemic mixture.

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E. Challenge

8. Butan-2-ol and butan-1-ol are both $C_4H_{10}O$. Explain why butan-2-ol has two stereoisomers but butan-1-ol has none, and name the type of isomerism that relates butan-1-ol to butan-2-ol.

F. CIP rules and R/S

9. Use the CIP rules to rank each set of groups from highest to lowest priority, and explain the tie-breaks: (a) –OH, –CH_3, –H, –Cl (b) –COOH, –CH_2OH, –CHO, –CH_3 (c) –CH_2CH_3, –$CH(CH_3)_2$, –CH_2OH

10. Lactic acid is $CH_3CH(OH)COOH$. (a) Rank the four groups on its chiral carbon. (b) In one drawing, H points away from you and, going clockwise, you meet OH, then COOH, then CH_3. Is it R or S? (c) In another drawing, H is on a solid wedge and the path $1 \rightarrow 2 \rightarrow 3$ looks clockwise. Is it R or S?

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